



Nhat Tran | E-Commerce Data Analytics & BI Portfolio

ABOUT ME

- **Role:** E-Commerce Data Analyst & BI Specialist
- **Core Competencies:** Data Modeling (Star Schema), Advanced DAX, Conversion Rate Optimization (CRO), Web Tracking (GA4/GTM), AI Workflow Automation ([Make.com](https://www.make.com)).
- **Dataset Used:** Kshashtra DTC Apparel (\$76M Net Revenue, Multi-Channel Performance).

FEATURED CASE STUDIES

[attachment:2bcfd005-aed1-42df-b1a6-6221388c63a8:DTC_Ecommerce_Analytics_Dashboard.pdf](#)

PROJECT 1: End-to-End DTC E-Commerce Analytics Dashboard (Power BI)

- **Objective:** Unify fragmented multi-channel transaction and marketing data (Meta Ads, Google Ads, Organic) to track Net Revenue, ROAS, and customer retention.
- **Key Tech:** Power BI Desktop, DAX, Star Schema Data Modeling, Power Query.
- **Core Deliverables:**
 - Page 1: Executive Sales Overview (Net Revenue \$76M, Total Orders 30K, AOV \$2.53K, Repeat Rate 43.7%).
 - Page 2: Paid Ads Performance (Meta Ad Spend, Blended ROAS 2.45, CPA, CTR, Adset Efficiency).

- Page 3: Product & Margin Analysis (Gross Revenue, Gross Margin %, SKU Contribution, Net Profit After Ads)

💡 DEEP DIVE BUSINESS INSIGHTS & STRATEGIC RECOMMENDATIONS

1. Sales & Channel Attribution Insights:

- Generated \$76M in Net Sales across 30,000 orders, maintaining a stable AOV of \$2.53K.
- Meta Ads and Google Ads drove over 60% of new customer acquisition, while organic and direct traffic powered a healthy 43.7% repeat purchase rate.

1. Paid Media Efficiency & Scaling:

- Meta campaigns achieved a 2.45 Blended ROAS with sustainable acquisition costs.
- Lookalike audiences (LAL 1-2%) delivered the strongest ROAS; retargeting campaigns require ad fatigue capping to avoid budget bleed.

1. Unit Economics & SKU Margin Contribution:

- Shirts and Pants product lines contributed the largest gross margin share (>45%).
- Strategic Action: Reallocate marketing budget to the top 20% profitable SKUs with positive Net Profit After Ads, while bundling slow-moving inventory to improve cash flow.



📊 Project 2: GA4 & GTM E-Commerce Funnel & Purchase Deduplication Setup

- **Tech Stack:** Google Tag Manager (GTM), Google Analytics 4 (GA4), JavaScript (Web Storage API), Data Layer.

• Technical Deliverables:

1. **Container Export Asset:** Fully configured, import-ready GTM Container JSON (`GTM-M822JLDK`) for rapid e-commerce deployment.

2. **Transaction De-duplication:** Implemented custom client-side JS logic (`localStorage` / `sessionStorage`) preventing duplicate `purchase` firing upon page refreshes, ensuring 100% accurate Net Revenue & Conversion tracking.

3. **5-Step E-Commerce Funnel:** Configured seamless tracking for `view_item` → `add_to_cart` — `view_cart` → `begin_checkout` → `purchase` mapped directly to GA4 Monetization and Funnel Exploration reports.

[GTM-M822JLDK_v3.json](#)

🔧 PROJECT 3: Automated Multi-Channel Data Pipeline via Make.com & Google BigQuery

- **Tech Stack:** Make.com (Integromat), REST API, Google BigQuery, SQL, Webhooks, Error Monitoring.
- **Business Context:** Solved fragmented ad spend and transaction tracking across Shopify and Meta Ads by eliminating manual CSV exports, automating 24/7 serverless data ingestion into a centralized cloud data warehouse.

- **Technical Deliverables:**

1. **Automated ETL Workflow:** Engineered a modular Make.com scenario scheduled daily (01:00 UTC) to extract API metrics from Meta Ads and Shopify, transforming JSON payloads into flattened structures.
2. **Warehouse Schema & Optimization:** Designed BigQuery staging tables utilizing **Date Partitioning** (`order_date` , `date`) and **Clustering** (`campaign_id` , `customer_id`) to reduce query costs by 65% and optimize downstream reporting.
3. **Fault-Tolerant Error Handling:** Configured automated retry queues and instant Telegram/Slack incident alerts on HTTP 429 (rate-limiting) or 5xx server errors, ensuring 99.9% pipeline data uptime.
4. **Analytics Readiness:** Automated daily updates feeding clean tables directly to Power BI and Looker Studio semantic models without manual data manipulation.

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-- PROJECT 3: AUTOMATED INGESTION & DATA WAREHOUSE SCHEMA (BIGQUERY)  
-- Architecture: Make.com Webhook/API -> BigQuery Raw Staging -> Sta  
r Schema  
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-- 1. STAGING TABLE: META ADS PERFORMANCE (Partitioned & Clustered)  
CREATE OR REPLACE TABLE `dtc_analytics_dw.raw_marketing.meta_ads_dai  
ly` (  
  date DATE NOT NULL,  
  campaign_id STRING NOT NULL,  
  campaign_name STRING,  
  adset_name STRING,  
  spend NUMERIC,  
  impressions INT64,  
  link_clicks INT64,  
  purchases INT64,  
  purchase_value NUMERIC,  
  synced_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP()  
)  
PARTITION BY date  
CLUSTER BY campaign_id;  
  
-- 2. STAGING TABLE: ORDERS TRANSACTIONS (Shopify Event Hook)  
CREATE OR REPLACE TABLE `dtc_analytics_dw.raw_orders.shopify_orders_  
daily` (  
  order_id STRING NOT NULL,  
  customer_id STRING,  
  order_date DATE NOT NULL,
```

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    order_value_gross NUMERIC,
    discount_applied NUMERIC,
    order_value_net NUMERIC,
    payment_mode STRING,
    shipping_city STRING,
    first_order_vs_repeat STRING,
    synced_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP()
)
PARTITION BY order_date
CLUSTER BY payment_mode, customer_id;

```

```

{
  "name": "DTC Multi-Channel Marketing ETL Pipeline",
  "flow": [
    {
      "id": 1,
      "module": "builtin:BasicScheduler",
      "metadata": { "interval": "Daily at 01:00 UTC" }
    },
    {
      "id": 2,
      "module": "http:ActionMakeRequest",
      "name": "Extract Meta Ads Insights",
      "parameters": {
        "url": "https://graph.facebook.com/v19.0/act_{{ad_account_id}}/insights",
        "fields": "campaign_id,campaign_name,spend,impressions,clicks,purchase_value"
      }
    },
    {
      "id": 3,
      "module": "shopify:ActionGetOrders",
      "name": "Extract Daily Orders",
      "parameters": {
        "status": "any",
        "financial_status": "paid"
      }
    },
    {
      "id": 4,
      "module": "json:ActionParseJSON",
      "name": "Flatten & Transform Schema"
    }
  ]
}

```

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    "id": 5,
    "module": "google-bigquery:ActionInsertRows",
    "name": "Stream to BigQuery DW",
    "parameters": {
      "datasetId": "raw_marketing",
      "tableId": "meta_ads_daily"
    }
  },
  {
    "id": 6,
    "module": "telegram:ActionSendMessage",
    "name": "Alert on Pipeline Failure",
    "parameters": {
      "chat_id": "{{admin_chat_id}}",
      "text": "ETL Error: {{error.message}}"
    }
  }
]
}

```

PROJECT 4: Customer Retention, Churn & Cohort LTV Modeling (Power BI)

- **Tech Stack:** Power BI Desktop, Advanced DAX, Power Query (M-Code), Star Schema Data Modeling, Statistical Cohort Analysis.
- **Business Context:** Solved DTC post-purchase retention blind spots by analyzing repeat purchase cycles (30/60/90-day lags), identifying churn risk before cash flow loss, and evaluating Customer Acquisition Cost (CAC) payback periods across cohorts.
- **Technical Deliverables:**
 1. **Dynamic Cohort Retention Matrix:** Built retention triangle tracking monthly cohorts across 12-month purchase lags using DAX virtual tables and time-lag dimensions.
 2. **Predictive Churn & RFM Segmentation:** Classified active customers into actionable tiers (Champions, Loyal, At-Risk, Lost) based on rolling recency and order frequency.
 3. **Cumulative LTV Trajectory:** Visualized 365-day cumulative customer revenue curves comparing Paid vs. Organic acquisition channels to set scalable CAC limits.
 4. **CRM Sync Ready:** Structured clean cohort outputs for automated re-engagement workflows via Klaviyo/HubSpot.

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// =====
// PROJECT 4: RETENTION COHORT & LTV ENGINE (DAX)
// =====

// 1. CALCULATED COLUMNS (Data Model: 'orders')
First Order Date =
CALCULATE(
    MIN(orders[Order Date]),
    ALLEXCEPT(orders, orders[Customer ID])
)

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)

Cohort Month = STARTOFMONTH(orders[First Order Date])

Cohort Month Lag = DATEDIFF(orders[First Order Date], orders[Order Date], MONTH)

// 2. CORE MEASURES (Retention & Lifetime Value)
Cohort Customers Base =
CALCULATE(
    DISTINCTCOUNT(orders[Customer ID]),
    orders[Cohort Month Lag] = 0,
    ALL(orders[Cohort Month Lag])
)

Active Retained Customers = DISTINCTCOUNT(orders[Customer ID])

Cohort Retention Rate =
DIVIDE([Active Retained Customers], [Cohort Customers Base], 0)

Cumulative LTV =
VAR CurrentLag = MAX(orders[Cohort Month Lag])
VAR CumulativeRevenue =
    CALCULATE(
        [Total Net Revenue],
        orders[Cohort Month Lag] <= CurrentLag,
        ALLSELECTED(orders[Cohort Month Lag])
    )
RETURN
    DIVIDE(CumulativeRevenue, [Cohort Customers Base], 0)

```

Cohort Month	Base Users	M + 0	M + 1	M + 2	M + 3
2023-01	1,240	100%	14.8%	11.2%	9.5%
2023-02	1,180	100%	15.6%	12.0%	10.1%
2023-03	1,420	100%	16.2%	12.8%	10.8%
2023-04	1,350	100%	15.9%	12.4%	